AN ADAPTIVE MOBILE LEARNING ENVIRONMENT FOR CHILDREN WITH DYSLEXIC LEARNING PROFILES.

25-26J-296

Project Proposal Report

Vithanage B H

BSc (Hons) Degree in Information Technology Specialized in Information Technology

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
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..... 27/08/2025

ABSTRACT

This project aims to create a Visual Processing Improvement Module within a mobile app to help children with dyslexia recognize colors better. It addresses their visual processing issues that affect reading, writing, and overall academic performance. Dyslexia often makes it hard to tell similar colors or shapes apart, leading to confusion with text and increased visual stress. The research focuses on using machine learning and image processing to design a personalized module that improves visual discrimination, comprehension, and learning outcomes. The study uses a modular system design, featuring a Flutter-based user interface for fun, kid-friendly activities, a Python-driven controller for machine learning, and a MongoDB backend for storing performance data and visual tasks. The module adjusts task difficulty and color combinations in real time, based on how well the user performs, like their accuracy in matching colors. This ensures customized support. Multimodal feedback combines visual, auditory, and tactile cues to boost engagement and help with retention of learning. The design emphasizes scalability, accessibility—like adjustable contrasts and language options—and data security to meet privacy rules. By filling the gaps left by static interventions and limited multimodal integration, this module is expected to greatly enhance visual processing skills. This improvement could lead to better handwriting fluency, increased reading endurance, and greater academic confidence in children with dyslexia. The findings will add to educational technology by showing how effective machine learning-driven, adaptive visual interventions can be, providing a scalable option for various educational environments.

Keywords: Dyslexia, Machine Learning, Flutter-based, Visual discrimination, Comprehension,

LIST OF TABLES

Table 1 List of Abbreviations

LIST OF APPENDIXES

Appendix A: Plagiarism Report

LIST OF ABBREVIATIONS

Key Word	Meaning
ML	Machine Learning
UI	User Interface
TTS	Text To Speech
GCP	Google Cloud Platform
COPPA	Children's Online Privacy Protection Act
GDPR	General Data Protection Regulation
API	Application Programming Interface
DB	Data Base

Table 1:List of Abbreviations

1. INTRODUCTION

Dyslexia is a learning disorder that affects 15-20% of people. It impacts reading, writing, and spelling, even when a person has normal intelligence [1][2][3]. A major challenge for many dyslexic children is trouble with visual processing, especially in identifying and distinguishing colors. This difficulty makes it hard for them to tell apart similar letters (like "b" and "d") or words (like "was" and "saw") [4][5]. These visual issues often relate to visual stress or low contrast sensitivity. They create text distortion, which reduces reading fluency and academic confidence [4][10]. This project suggests a Visual Processing Improvement Module in a mobile app that focuses on improving color identification to tackle these problems. By using machine learning and image processing, the module offers engaging, game-like activities (such as color matching and overlay customization) that adapt to a child's performance. This can enhance their visual discrimination and comprehension.

When we look at background visual processing difficulties in dyslexia, like sensitivity to glare or poor color differentiation, these issues prevent accurate text perception and lead to reading mistakes and fatigue [6]. Traditional methods, such as colored overlays, provide some help but are not personalized [6][10]. Recent studies, including Tzovaras et al. (2018), show that image processing can improve visual skills by customizing stimuli. Brown et al. (2019) also point out that machine learning can adapt tasks in real-time, which boosts engagement [7][8]. Mobile apps can serve as accessible and engaging platforms for these interventions [12]. This module uses machine learning to customize color-based tasks and combines various types of feedback (visual, auditory, and tactile) to enhance visual processing, handwriting fluency, and academic success for children with dyslexia.

1.1Background and Literature Survey

Dyslexia is a well-known learning disorder that affects 15-20% of people, especially children. It causes difficulties in reading, writing, and spelling, even though these individuals have normal intelligence. A major issue for many children with dyslexia is their struggle with visual processing, particularly in identifying and distinguishing colors. This issue shows up as trouble telling apart similar letters, like "b" and "d," or words, such as "was" and "saw." These difficulties are often linked to visual stress or low contrast sensitivity. They can cause text to appear distorted, slow down reading speed, and lower academic confidence. Addressing color identification is crucial because it directly affects the ability to process visual information needed for schoolwork.

Recent advancements in educational technology show promise for technology-based interventions that can improve color identification in children with dyslexia. Traditional methods, like colored overlays, help reduce visual stress but lack personalization and flexibility. Research by Tzovaras et al. (2018) revealed that image processing techniques can improve visual abilities by adjusting visual stimuli based on individual performance. This helps boost discrimination skills. Similarly, Brown et al. (2019) looked into using machine learning algorithms to create personalized learning experiences. These algorithms can change task difficulty in real time based on how a child progresses, leading to significant gains in visual processing. Multimodal approaches that use visual, auditory, and tactile feedback can further improve engagement and learning results by utilizing multiple senses. Mobile apps, with their accessibility and engaging interfaces, are perfect for implementing these interventions, making learning fun and scalable. However, existing solutions often do not provide real-time adaptability or thorough multimodal integration. This creates an opportunity to develop a more effective, machine learning-driven module to improve color identification and visual processing in children with dyslexia.

1.2Research Gap

Despite significant progress in educational technology, a noticeable gap remains in helping children with dyslexia who struggle with color identification and visual processing. Current interventions have important limitations:

- ✓ Lack of Dynamic Adaptability: Many existing solutions, like static colored overlays, provide fixed visual stimuli that do not change in real time based on a child's performance or needs. While some tools allow teachers to make manual changes, few use machine learning to adjust color-based tasks dynamically. This limits the development of a truly personalized learning environment that grows with the child's visual processing skills [9].
- ✓ Limited Multimodal Integration: Research shows that using multiple senses—visual, auditory, and tactile—improves learning for children with dyslexia. However, most tools focus only on visual stimuli and miss the chance to include auditory or tactile cues, which could help improve understanding and memory through multisensory engagement [10].
- ✓ Inadequate Performance Tracking and Feedback: Current solutions often lack strong real-time performance monitoring and smart data analysis. Few interventions provide useful insights or adapt color combinations on the fly, which makes it harder to keep dyslexic learners engaged and motivated [11][12].

- ✓ Scalability and Accessibility Issues: Many existing tools do not cater to diverse dyslexic profiles or different educational contexts. They often lack features like adjustable interfaces or language support, limiting their scalability and accessibility.

These gaps highlight the need for a solution that is adaptable, uses multiple senses, and can scale to improve color identification for children with dyslexia.

1.3Research Problem

Research problem statement
How can machine learning and image processing algorithms be used to create a Visual Processing Improvement Module that adjusts color identification activities and games? This module aims to improve visual discrimination, comprehension, and overall learning for children with dyslexia.
To address this problem, the research will focus on developing a module that:

1. Dynamically Adapts to Individual Needs
   - ✓ The module will use machine learning to monitor performance data, such as accuracy in color matching, and adjust task complexity and color combinations in real-time. This ensures a personalized learning experience based on each child's visual processing strengths and weaknesses [14].

2. Incorporates Multimodal Feedback
   - ✓ Visual, auditory, and tactile feedback will support learning. For instance, audio cues will be paired with color-based tasks to reinforce discrimination and retention [15].

3. Provides Robust Performance Tracking and Feedback
   - ✓ The module will track progress and generate insights to customize interventions. It will provide useful feedback to educators and learners [16].

4. Ensures Scalability and Accessibility
   - ✓ The solution will be scalable across educational contexts and have accessible features like adjustable interfaces to support various dyslexic profiles.

2. OBJECTIVES

2.1 Main Objectives

The main goal of this project is to create a Visual Processing Improvement Module designed to improve color identification and visual processing skills in children with dyslexia. Dyslexia is a learning disorder that affects reading, writing, and spelling. It often arises from difficulties in visual discrimination and processing, such as telling apart similar colors, letters, or patterns. These difficulties can cause confusion with text and create visual stress. Such challenges can hurt academic performance and self-confidence, especially in tasks that require precise color perception and visual-motor integration.

The proposed module uses modern technologies such as machine learning and image processing to create a personalized learning environment. It adjusts color-based tasks, like matching or selecting overlays, based on each child's performance. This way, activities stay appropriately challenging and avoid causing frustration or disengagement. This adaptability helps kids continually improve their visual processing skills. The module aims to enhance color identification, visual discrimination, and comprehension. Ultimately, it seeks to boost reading fluency, handwriting, and academic confidence in children with dyslexia. By tackling these essential visual processing issues, the module aims to offer an engaging, scalable solution for various learning settings within educational technology.

2.2 Specific Objectives

The following specific goals aim to achieve the main objective of developing a Visual Processing Improvement Module that helps children with dyslexia identify colors

- ✓ Improve Visual Discrimination and Comprehension: Provide exercises that help kids tell apart similar visual stimuli, such as colors, letters, and shapes. This will improve their recognition and understanding of visual information.
- ✓ Implement Dynamic Difficulty Adjustment: Use machine learning to change the difficulty of color identification tasks in real-time. This will be based on ongoing performance monitoring, ensuring each child is challenged just right without feeling overwhelmed.

- ✓ Integrate Multimodal Feedback Mechanisms: Add visual, auditory, and tactile feedback to color-based activities. This will boost engagement and cater to different sensory preferences, making learning more effective and interactive.
- ✓ Provide Performance Tracking and Analytics: Create a solid system to monitor progress in color identification tasks. It should generate detailed insights to guide adjustments and offer useful feedback to parents and educators.
- ✓ Ensure Scalability and Accessibility: Design a module that can be scaled to fit various educational settings. It should include features like adjustable color contrasts and language support to meet the needs of different dyslexic profiles.
- ✓ Develop Engaging and Interactive Learning Activities: Create a mix of games, quizzes, and exercises centered on color identification. This will help keep motivation high and encourage continuous improvement in visual processing skills.

3.METHODOLOGY

3.1 Complete System Architecture

The Visual Processing Improvement Module for better color identification will be developed using a modular and scalable design. This will support flexible functionality within a mobile application. The system has three main layers to provide fun and personalized color-based activities for dyslexic children.

User Interface (UI) Layer

This layer allows the dyslexic child to interact with the system. It is developed using Flutter, which ensures compatibility with mobile devices and tablets. The UI will offer kid-friendly activities for color identification, such as color matching games, overlay customization, and pattern recognition quizzes. These activities provide real-time feedback to keep children engaged. Accessibility features include adjustable color contrasts and dyslexia-friendly fonts to meet various visual processing needs.

Controller Layer

As the central hub, the controller layer manages the UI, backend services, and databases. It is built with Python and integrates machine learning using TensorFlow or PyTorch and image processing using OpenCV. This combination analyzes performance data, like

accuracy in color matching, and changes task difficulty and color combinations accordingly. This ensures a responsive experience that adapts to each child's visual processing abilities.

Backend Services and Database Layer

This layer handles data storage and processing, using MongoDB via pymongo for flexibility. It includes:

- ✓ Color Activities Database: Stores exercises such as color sorting and contrast puzzles.
- ✓ Machine Learning Engine Database: Holds performance metrics for real-time task adjustment.
- ✓ Image Processing Database: Manages data for color recognition tasks.
- ✓ Quiz and Game Database: Contains interactive games to improve visual discrimination.

The backend supports secure data management and scalability, allowing the module to fit into different educational settings while meeting privacy regulations.

This architecture creates a smooth, adaptable, and engaging experience. It uses machine learning and image processing to help improve color identification and visual processing skills in dyslexic children.

3.2 Component Architecture

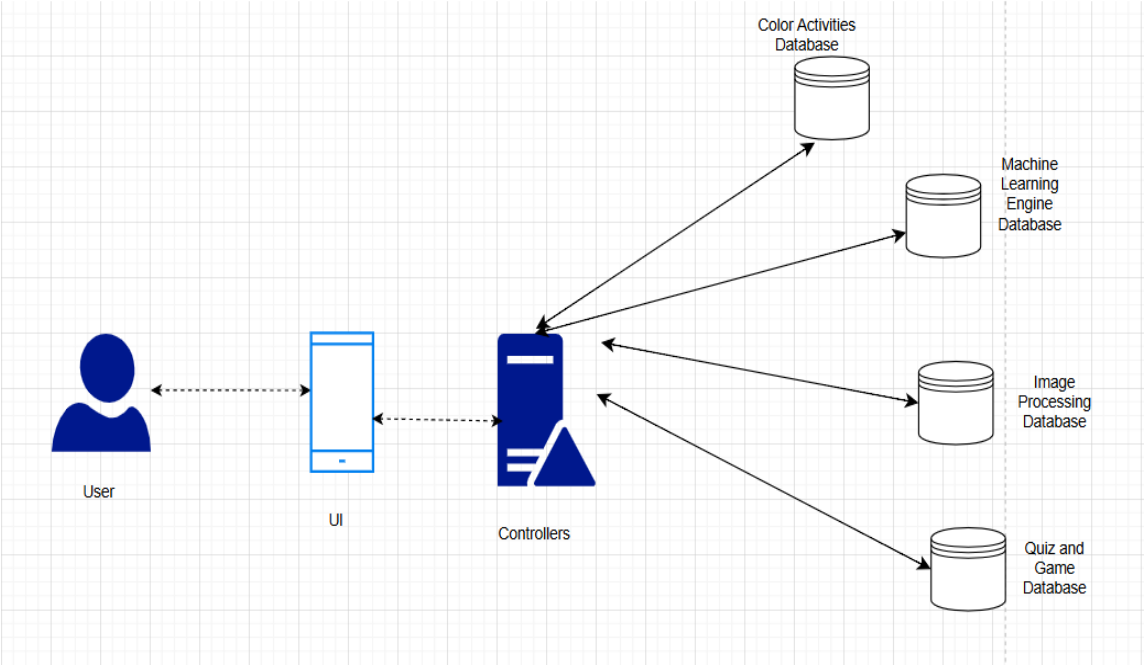

Figure 3.2 : Component Architecture

3.3 Functional Requirements

The Visual Processing Improvement Module for improving color identification in dyslexic children includes the following features to ensure effective, flexible, and engaging sessions:

- ✓ Dynamic Difficulty Adjustment: Automatically change the complexity of color identification tasks, like shade matching and contrast puzzles, in real-time based on the child's performance. It uses machine learning to tailor challenges to individual visual processing abilities.
- ✓ Color Activity Interaction: Offer a range of interactive exercises, such as color matching games, overlay customization, and pattern recognition, to help enhance visual discrimination and understanding.
- ✓ Performance Tracking: Track and assess progress across color-based tasks, generating detailed reports for children and educators to monitor improvements and pinpoint areas for growth.
- ✓ Multimodal Feedback: Provide feedback through visual, auditory, and tactile channels, such as audio cues for correct color matches and vibrations for task completion, to support learning and engagement.

- ✓ User Data Management: Safely store and handle performance metrics, interaction history, and personalized settings, ensuring data privacy and compliance with laws.

3.4 Non-Functional Requirements

The module will meet the following performance criteria to ensure reliability and accessibility:

- ✓ Reliability: Ensure minimal downtime and consistent performance. Users should have uninterrupted access to color identification activities.
- ✓ Scalability: Handle increased user loads and data efficiently. Support deployment in different educational settings without performance issues.
- ✓ Accessibility: Include features such as adjustable color contrasts, dyslexia-friendly fonts, and language support. This will help children with different levels of dyslexia and from various cultural backgrounds.
- ✓ Security: Use strong measures to protect user data. This includes performance metrics and interaction logs while following privacy rules like COPPA and GDPR.

3.5 Technology Selection

1. Programming Languages

- ✓ Python, chosen for backend development because of its wide range of libraries for machine learning, such as TensorFlow and PyTorch, and for image processing, like OpenCV.
- ✓ Dart (Flutter), used for frontend development to create a responsive, cross-platform user interface.

2. Databases

- ✓ MongoDB (pymongo), a NoSQL database selected for its ability to scale and its flexibility in managing unstructured data, including user interactions and performance metrics.

3. Machine Learning Frameworks

- ✓ TensorFlow/PyTorch, these frameworks will be used to build and deploy machine learning models that power the adaptive learning component.

4. Cloud Services

- ✓ Google Cloud Platform (GCP), selected for deploying the system because of its scalability, security, and built-in machine learning tools.

5. API Services

- ✓ Text-to-Speech and Speech Recognition APIs, Google's or other cloud-based APIs will be integrated to offer auditory feedback and enable voice interaction.

3.6 Ghant Chart

Figure 3.6: Ghant Chart

3.7 Work Breakdown Structure

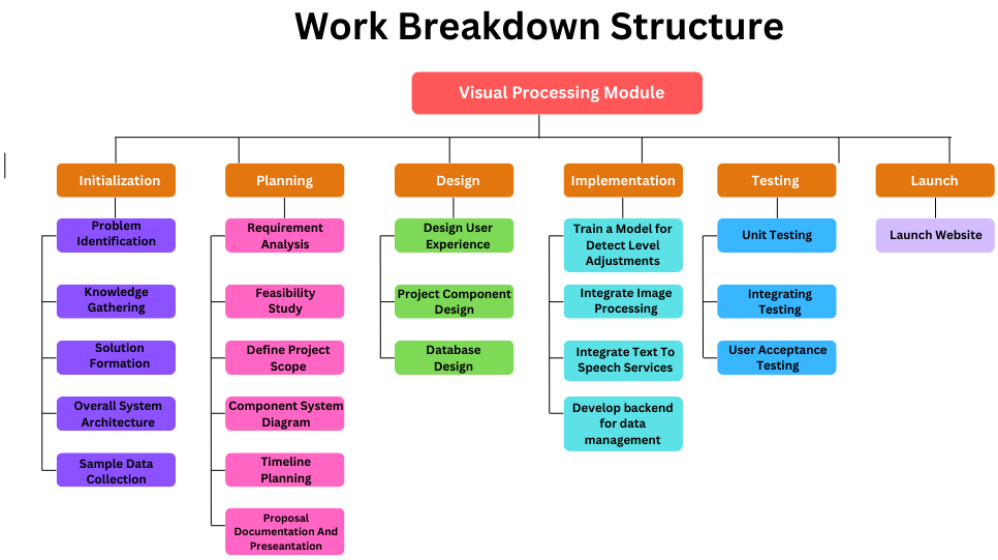

Figure 3.7: Breakdown Structure

3.8 User Requirements

When designing the mobile interface for children with dyslexia who want to improve their color identification skills, it is important to meet their specific visual processing needs. The interface should be easy to use, engaging, and simple. It needs to employ clear language and visuals that are suitable for their age. This will help guide users through activities related to color and provide feedback while avoiding complicated instructions. The system should motivate users to engage regularly, which can help improve visual discrimination. Additionally, it should cater to young users with different levels of dyslexia. Features like adjustable color contrasts and various feedback methods should be included.

Expected Use Cases

Use Case 1: Engage in Color Identification Activity

• **Actor:** Child with Dyslexia

• **Description:** The user plays a color matching game to enhance visual discrimination by identifying and matching similar shades or objects.

• **Preconditions:** The user has access to a compatible mobile device such as a smartphone or tablet and is logged into the app.

• **Trigger:** The user navigates to the color identification activity section and selects a game.

• **Basic Flow:**

1. The user opens the app and chooses a color matching game (e.g., “Match all blue items”).

2. The system displays a series of colored objects or shades on the screen.

3. The user selects the matching colors by touching the screen.

4. The system uses machine learning models to evaluate the choices for accuracy.

5. The system adjusts the difficulty of the task based on performance and offers feedback through visuals and sounds.

• **Postconditions:** The user finishes the activity and receives a summary of their progress, including performance metrics.

• **Exceptions:** Incorrect selections prompt encouraging feedback; if the system fails, users receive an option to try again.

Use Case 2: Receive Personalized Color Adjustment Recommendations

• **Actor:** Child with Dyslexia

• **Description:** The system offers customized suggestions for color settings, like text overlays, based on the user’s performance in color identification tasks.

• **Preconditions:** The user has completed at least one color identification activity, with data recorded on performance.

• **Trigger:** The system completes the performance analysis and turns on the recommendation engine.

• **Basic Flow:**

1. The system reviews performance metrics, such as accuracy and time taken to complete tasks.

2. The recommendation engine, powered by machine learning, creates personalized color settings (e.g., the best contrast for text overlays).

3. The user sees the suggestions on the interface (e.g., “Try a yellow background for reading”).

• **Postconditions:** The user applies the suggested settings to improve visual processing.

• **Exceptions:** If there is not enough performance data, no recommendations are generated, and the user is prompted to complete more tasks.

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APPENDICES Appendix

> Research Paper Checking

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Appendix A: Plagiarism Report